

Electric Vehicle Sales Analysis

Presented by :
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Problem Statement

AtliQ Motors is an automotive giant from the USA specializing in electric vehicles (EVs). In the last 5 years, their market share rose to 25% in the electric and hybrid vehicles segment in North America. As a part of their expansion plans, they wanted to launch their bestselling models in India, where their market share is less than 2%.

Bruce Haryali, the chief of AtliQ Motors India, wanted to do a detailed market study of the existing EV/Hybrid market in India before proceeding further. Bruce gave this task to the data analytics team of AtliQ Motors, and Peter Pandey is the data analyst working in this team.



Dataset Description

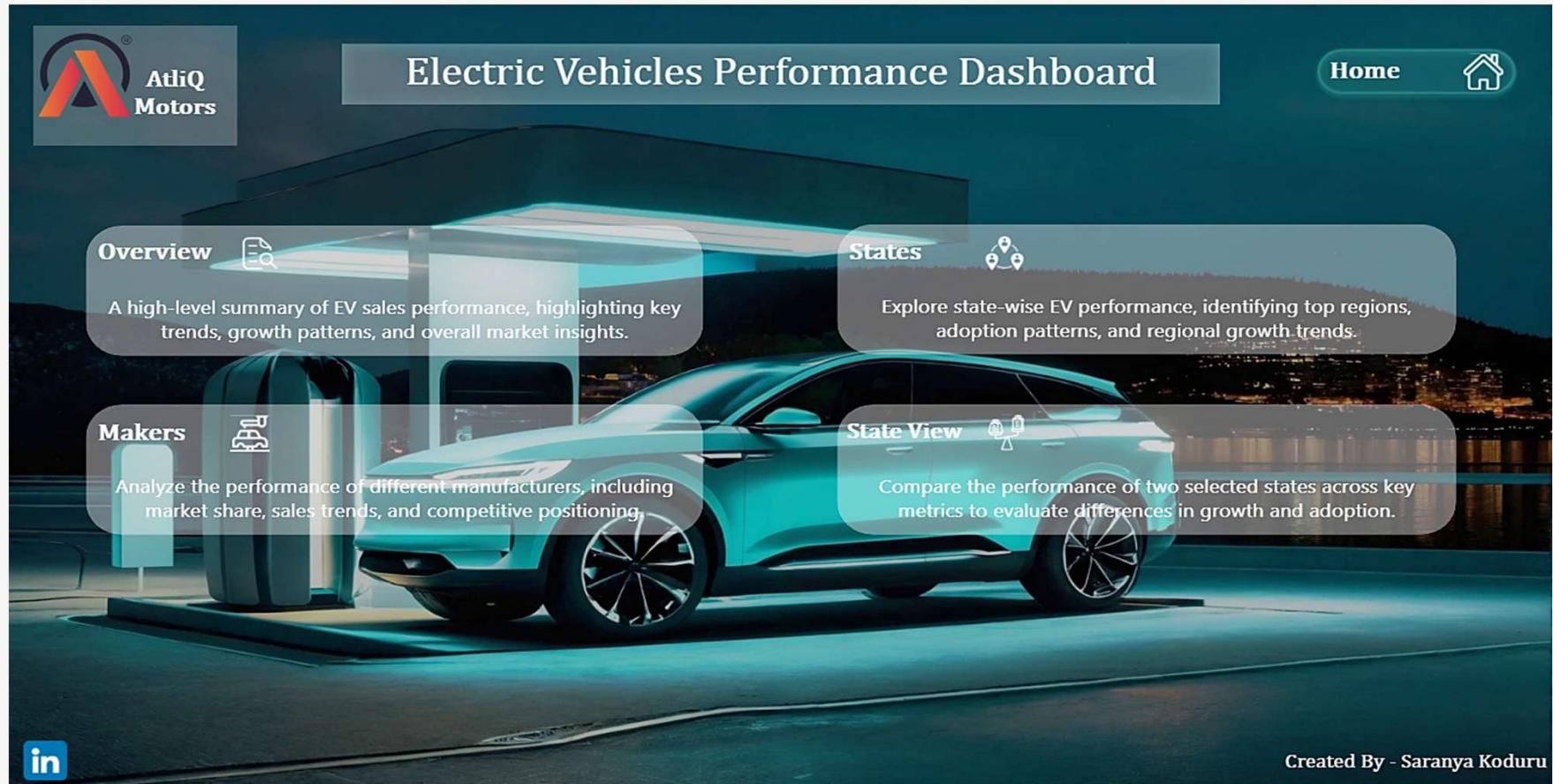
The datasets we have are –

- **electric_vehicle_sales_by_state**
renamed as: **state_sales**
date: Monthly record date (Format: DD-MMM-YY)
state: State name (location)
vehicle_category: 2-Wheeler or 4-Wheeler
electric_vehicles_sold: EV units sold
total_vehicles_sold: Total vehicles sold (EV + non-EV)
- **electric_vehicle_sales_by_makers**
renamed as: **maker_sales**
date: Monthly record date (Format: DD-MMM-YY)
vehicle_category: 2-Wheeler or 4-Wheeler
maker: Manufacturer/brand name
electric_vehicles_sold: EV units sold by maker
- **dim_date**
date: Monthly record date (Format: DD-MMM-YY)
fiscal_year: Financial year
quarter: Fiscal quarter (Q1–Q4)
(note : fiscal year starts on April 1st and ending on March 31st of the following year in India.)

Additional dimension tables were created to enhance data structure and analysis. They are –

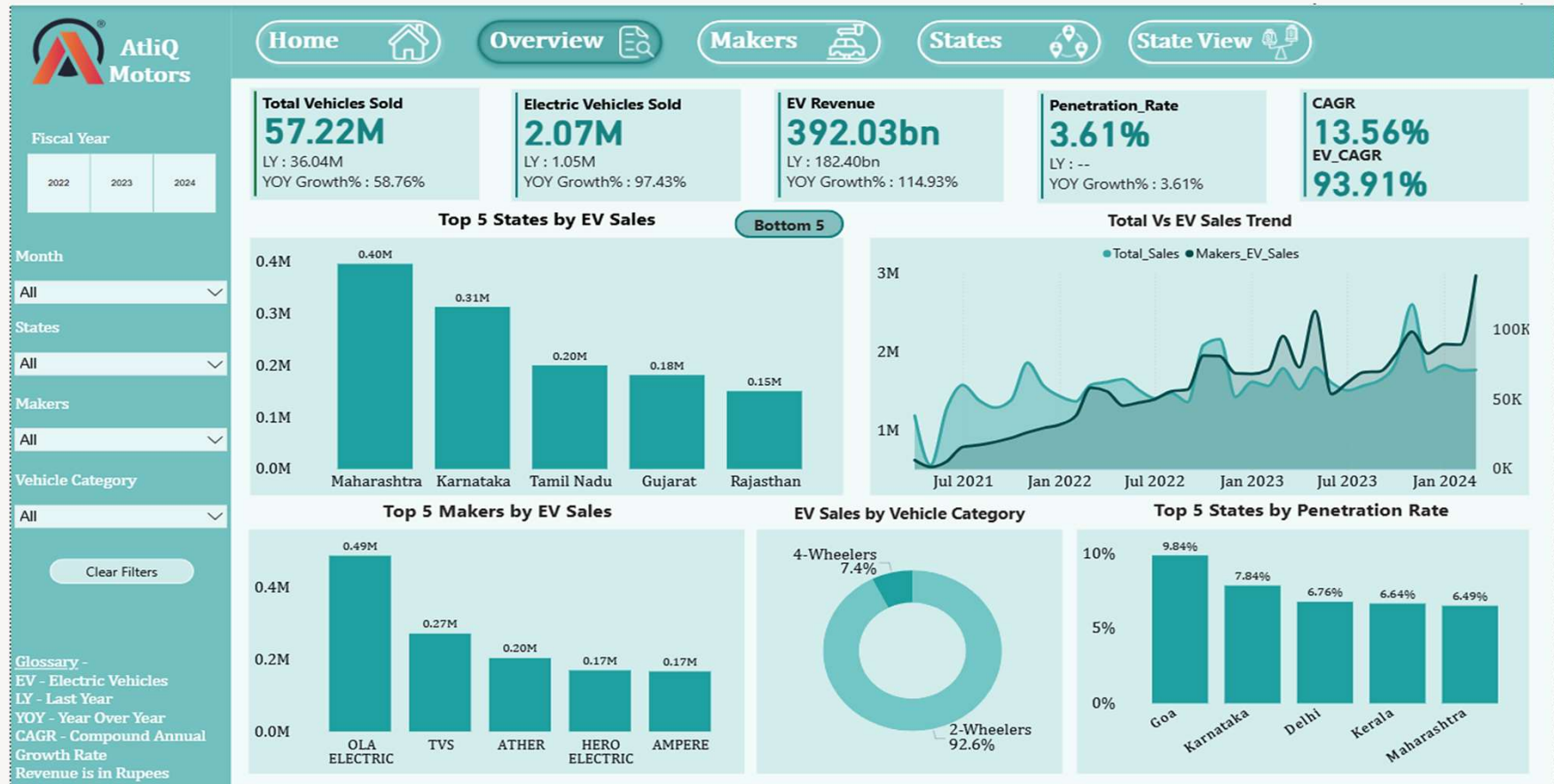
- **dim_state**
State/UT: state/ut name of India
- **dim_vehicle**
avg_price: average price set for each vehicle category
vehicle_category: 2-wheeler or 4-wheeler
- **dim_maker**
maker: maker name
- **dim_charging_stations**
charging_stations: Total charging stations for each state
State/UT: state/UT name

Dashboard – Home Page



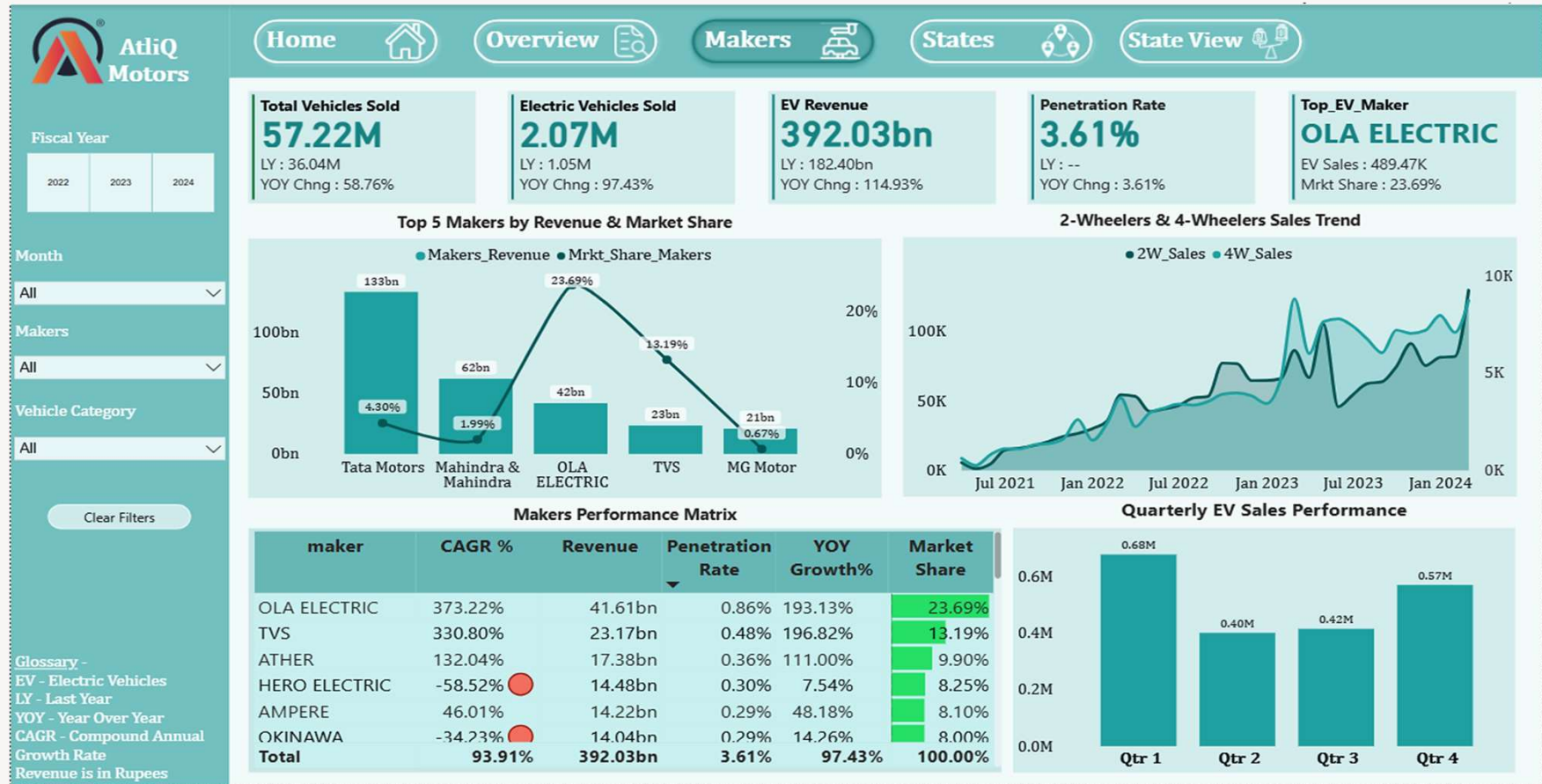
Dashboard – Overview Page

A quick snapshot of EV performance covering top states, sales trends, key players, category insights & overall KPIs.



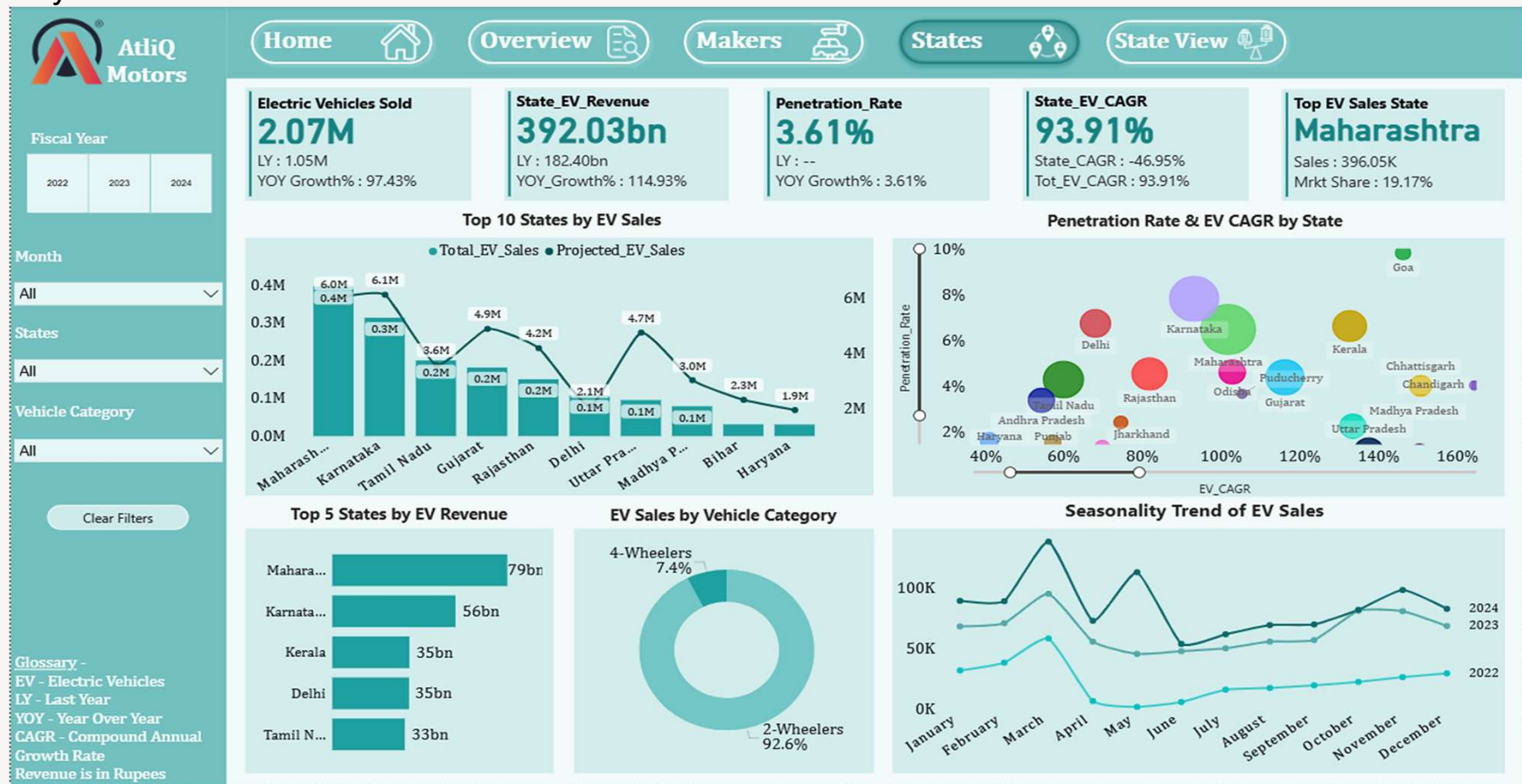
Dashboard – Makers Page

Concise view of EV makers, highlighting top performers, sales trends, growth metrics, and quarterly performance.



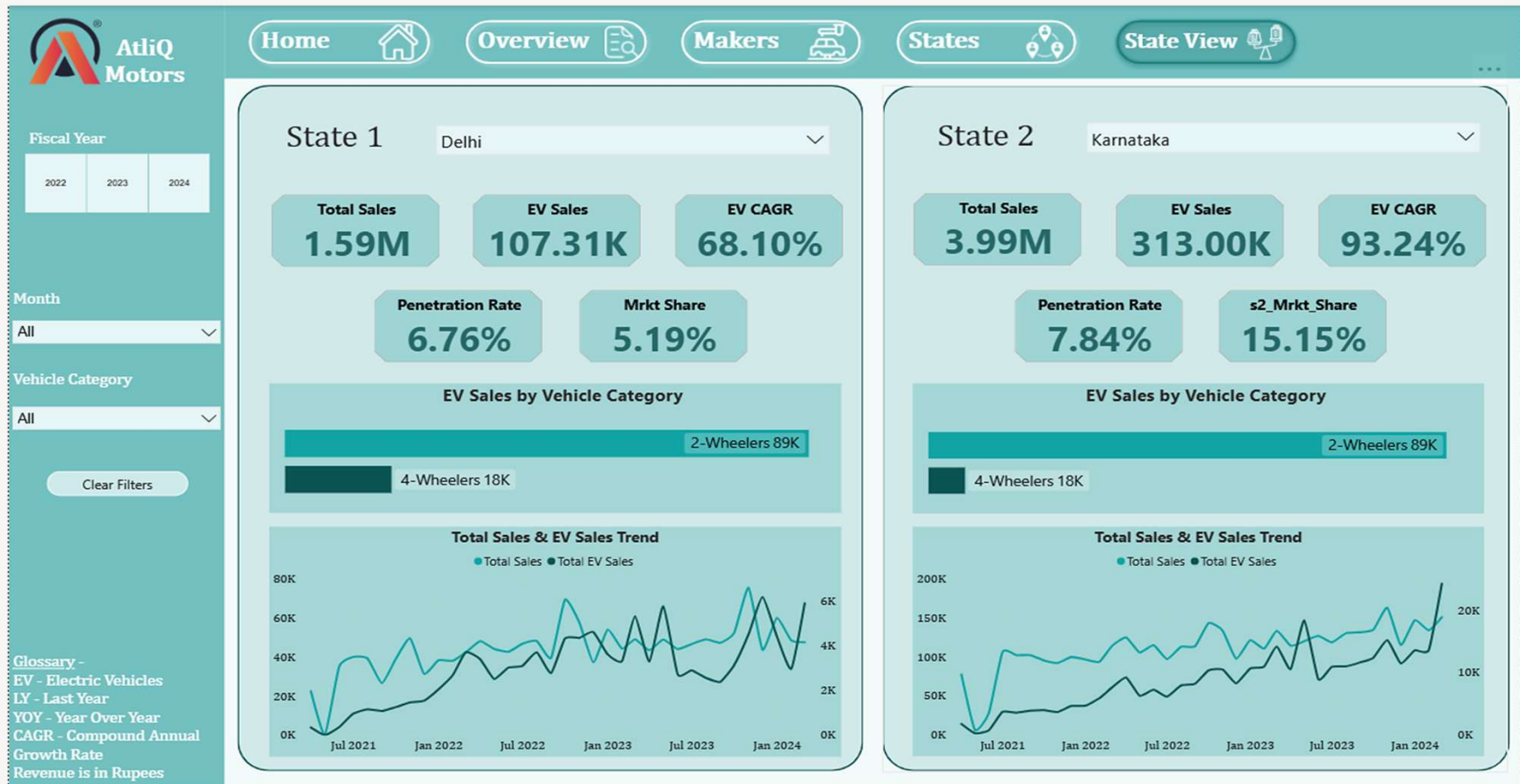
Dashboard – States Page

Detailed view of state-wise EV performance, showcasing top states, sales trends, revenue, vehicle categories, and key KPIs.



Dashboard – State View Page

Detailed comparison of total and EV sales, growth metrics, market share, and category-wise trends between two states.

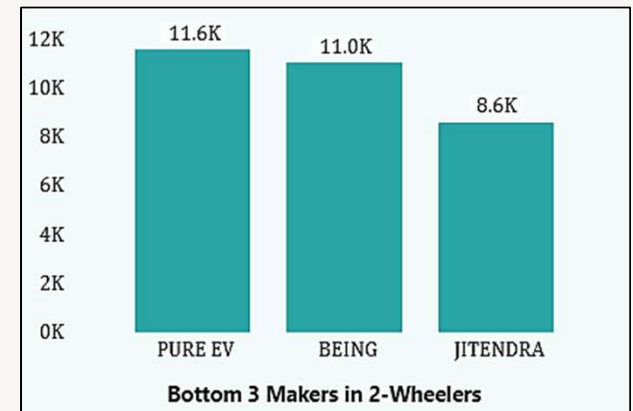
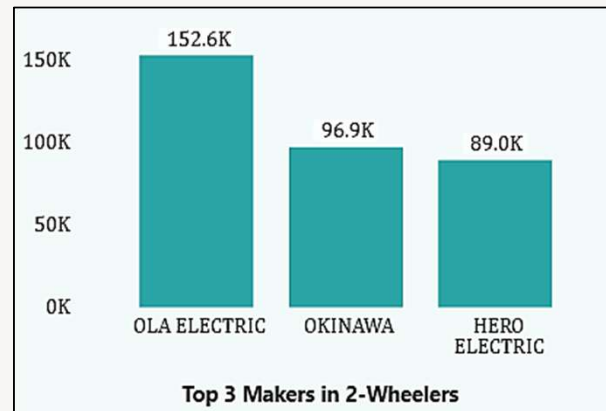


Preliminary Research Questions

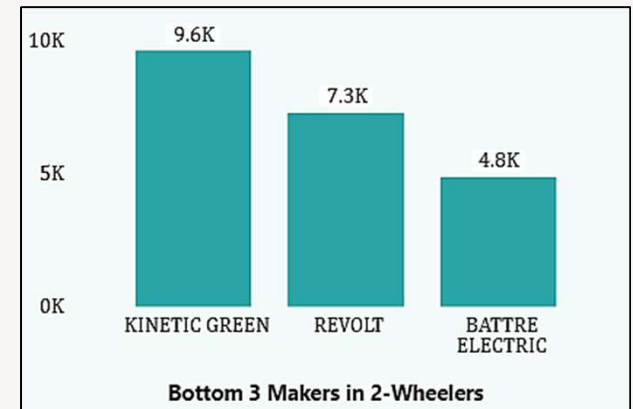
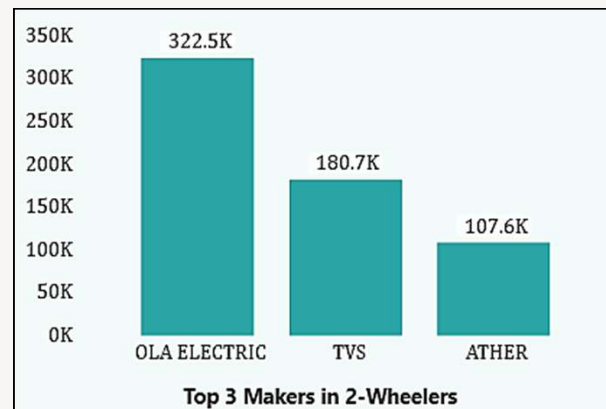
1. List the top 3 and bottom 3 makers for the fiscal years 2023 and 2024 in terms of the number of 2-wheelers sold.

2023

Ola Electric has consistently maintained its leadership in the 2W EV segment, demonstrating strong year-over-year growth.



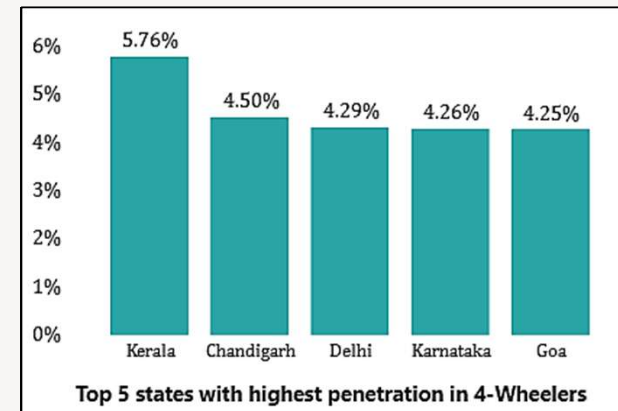
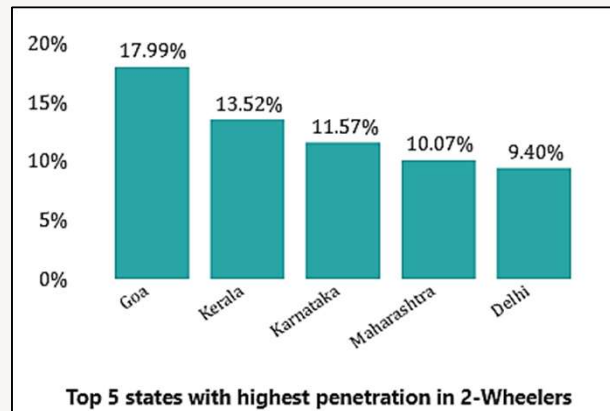
2024



Preliminary Research Questions

2. Identify the top 5 states with the highest penetration rate in 2-wheeler and 4-wheeler EV sales in FY 2024.

2024



3. List the states with negative penetration (decline) in EV sales from 2022 to 2024?

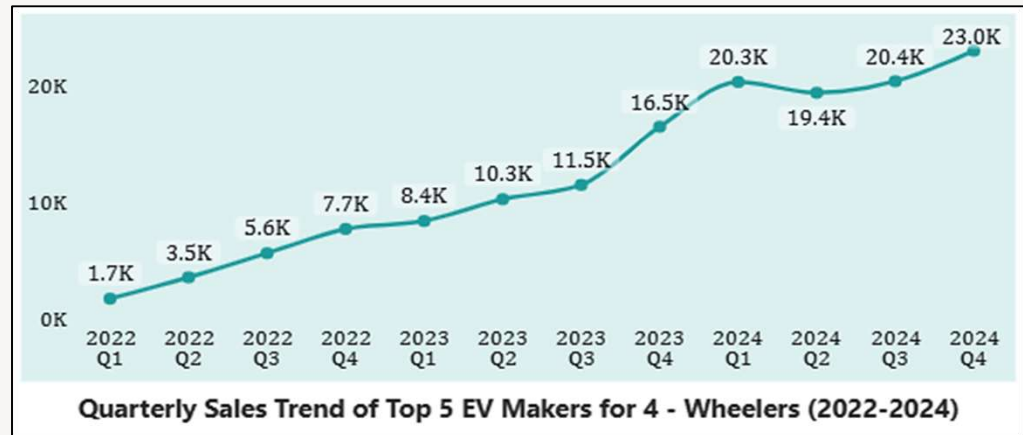
The table on the right presents penetration change from 2022 to 2024 in ascending order. As all values are positive, it indicates there is no decline in penetration and EV adoption has increased across all states in India. However, the relatively small magnitude of change suggests that growth remains gradual rather than rapid.

state/UT	Penetration_Chng
Sikkim	0.00%
Nagaland	0.05%
Andaman & Nicobar	0.06%
Arunachal Pradesh	0.11%
Meghalaya	0.35%
Himachal Pradesh	0.44%
Assam	0.45%

Preliminary Research Questions

4. What are the quarterly trends based on sales volume for the top 5 EV makers (4-wheelers) from 2022 to 2024?

The quarterly sales trend for 4-wheelers shows a consistent increase from 1.7K to 23K, indicating that EV sales have grown by over 13x between 2022 and 2024.



5. How do the EV sales and penetration rates in Delhi compare to Karnataka for 2024?

2024

Delhi		Karnataka	
Total_EV_Sales	Penetration_Rate	Total_EV_Sales	Penetration_Rate
46,724.00	7.71%	1,60,989.00	10.18%

In 2024, Karnataka outperforms Delhi in EV adoption, with a penetration rate of 10.18% compared to Delhi - 7.71%, reflecting a higher level of EV acceptance in the state.

Preliminary Research Questions

6. List down the compounded annual growth rate (CAGR) in 4-wheeler units for the top 5 makers from 2022 to 2024.

Maker	EV_Sales_22	EV_Sales_24	CAGR %
BYD India	33	1,466	566.52%
Hyundai Motor	110	1,390	255.48%
Mahindra & Mahindra	4,042	23,346	140.33%
MG Motor	1,647	8,829	131.53%
Tata Motors	12,708	48,181	94.71%
Total	18,540	83,212	111.85%

BYD India shows 566.5% CAGR, indicating strong growth potential, even though its total sales are still small.

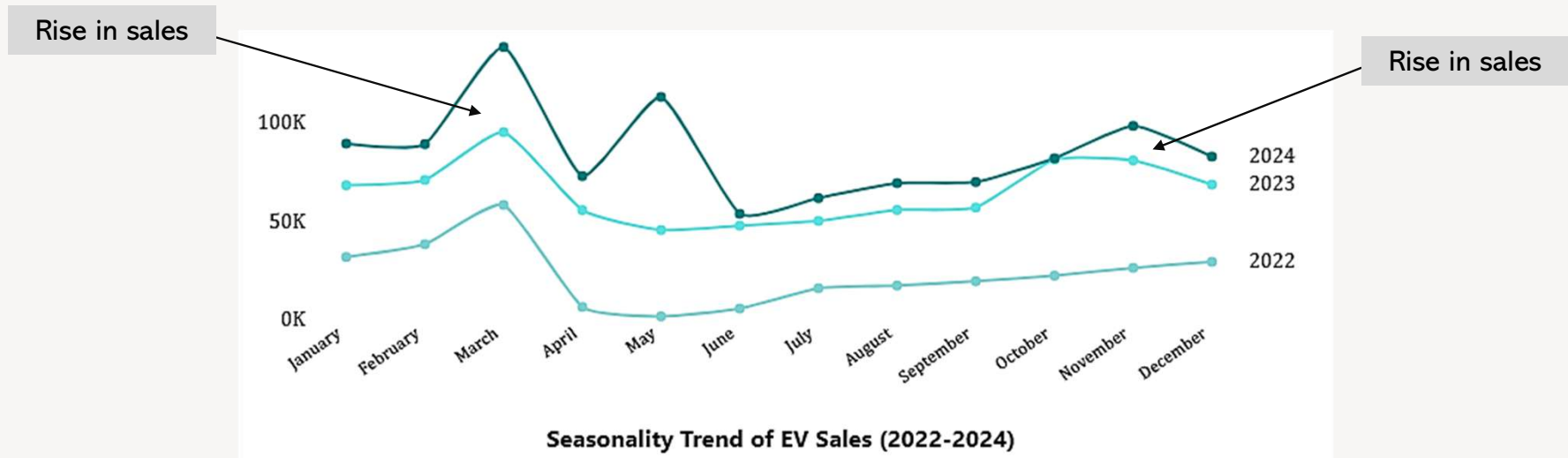
7. List down the top 10 states that had the highest compounded annual growth rate (CAGR) from 2022 to 2024 in total vehicles sold.

state/UT	TV_CAGR%
Meghalaya	28.47%
Goa	27.41%
Karnataka	25.28%
Delhi	22.88%
Rajasthan	21.50%
Gujarat	20.55%
Assam	20.13%
Mizoram	18.77%
Arunachal Pradesh	18.30%
Andaman & Nicobar	18.29%
Total	22.32%

Meghalaya shows highest CAGR indicating fastest growth momentum.

Preliminary Research Questions

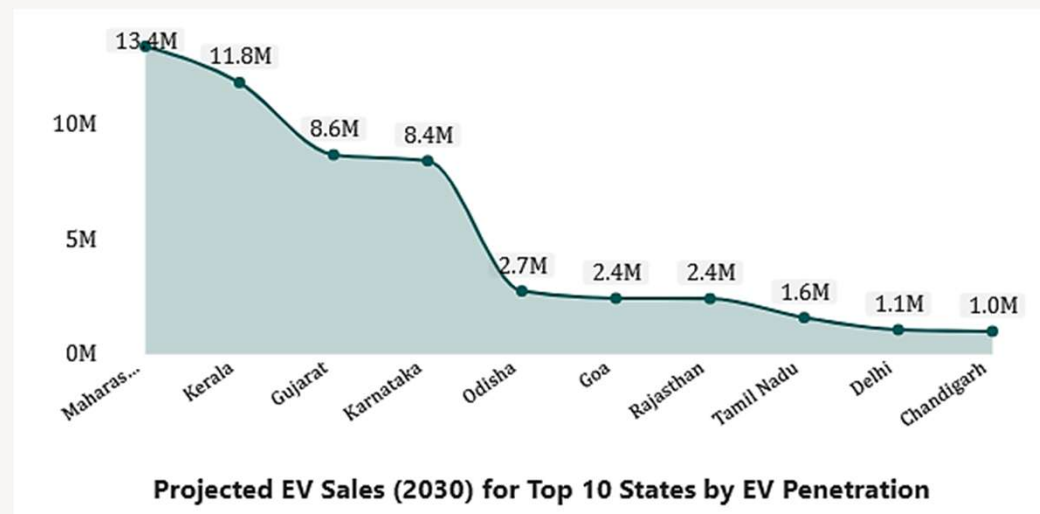
8. What are the peak and low season months for EV sales based on the data from 2022 to 2024?



If we look at the overall trend, sales start off strong and peak during February–March, mainly driven by year-end discounts and incentive-driven purchases. This is followed by a sharp drop in April–May, indicating the impact of reduced offers and a post-peak slowdown. From June onwards, sales gradually recover as demand picks up, and then rise again during October–November, supported by festive season demand and promotional offers.

Preliminary Research Questions

9. What is the projected number of EV sales (including 2-wheelers and 4-wheelers) for the top 10 states by penetration rate in 2030, based on the compounded annual growth rate (CAGR) from previous years?

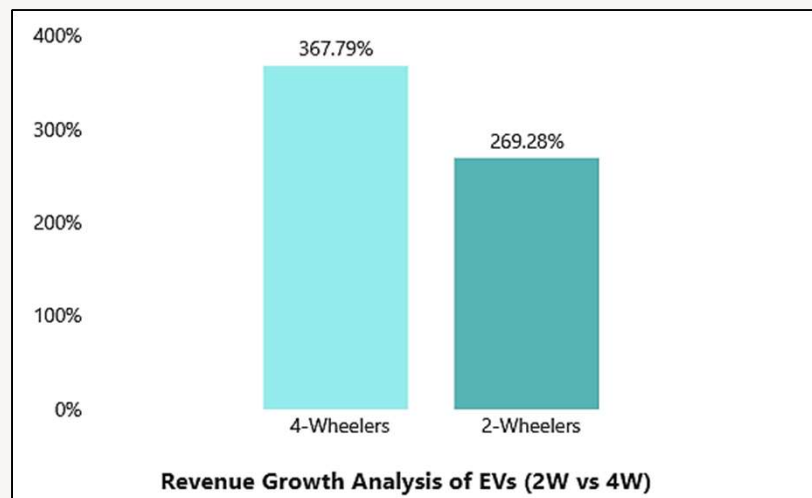


Based on the projected sales, Maharashtra ranks first with an estimated 13.4 million EV sales.

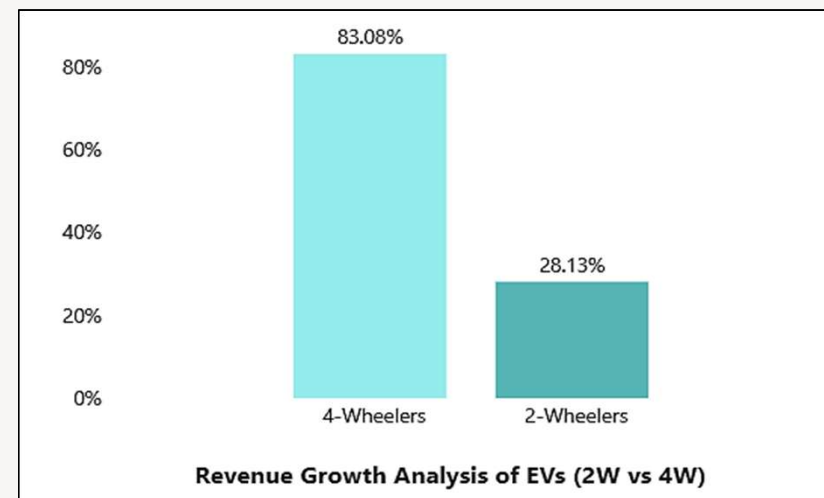
Preliminary Research Questions

10. Estimate the revenue growth rate of 4-wheeler and 2-wheelers EVs in India for 2022 vs 2024 and 2023 vs 2024, assuming an average unit price of ₹85,000.00 for 2-Wheelers and ₹15,00,000.00 for 4 – Wheelers.

2022 Vs 2024



2023 Vs 2024



Both 2W and 4W segments show strong overall growth (2022–2024), with 4W outperforming 2W. However, in the recent period (2023–2024), 2W growth has slowed significantly (28.13%) which shows it is becoming stable, whereas 4W continues to show strong YoY growth (83%) indicating strong demand and expansion.

Secondary Research Questions

1. What are the primary reasons for customers choosing 4-wheeler EVs in 2023 and 2024 (cost savings, environmental concerns, government incentives)?

Cost Savings (Key Driver)



- Lower Running Cost: EVs have much lower cost per km due to cheaper electricity vs petrol/diesel
- Low Maintenance: Fewer components → less servicing and lower maintenance cost
- Better Long-Term Value (TCO): Higher upfront cost, but more economical over time
- Fuel Price Impact: Rising fuel prices increased the financial appeal of EVs

Government Incentives



- FAME II Scheme: Provides subsidies and expands charging infrastructure, making EVs more accessible, affordable
- PLI Scheme: Promotes domestic battery manufacturing, reduces import dependency, and attracts investments
- Financial Incentives: Provides purchase subsidies, interest support & loans reduce upfront cost and improve affordability
- Tax & Convenience Benefits: Road tax exemptions and battery swapping improve cost efficiency and usability

Environmental Concerns



- Zero Emissions: EVs produce no tailpipe emissions, improving urban air quality and reducing greenhouse gases
- Reduced Noise Pollution: Quieter operation compared to petrol/diesel cars enhances urban living conditions
- Energy Efficiency: EVs use electricity more efficiently than ICE vehicles, lowering overall energy consumption per km
- Sustainable & Eco-Friendly Lifestyle: Awareness of climate change drives low-carbon vehicle adoption

Technology, Infrastructure & Market Growth



- Improved Battery Performance: Better efficiency and durability have increased EV reliability
- Extended Driving Range: Modern EVs offer competitive range, reducing range anxiety
- Charging Infrastructure: Growing number of public stations improves usability and convenience
- Market Growth & Trust: More models from Tata & MG and better service networks boost confidence

Secondary Research Questions

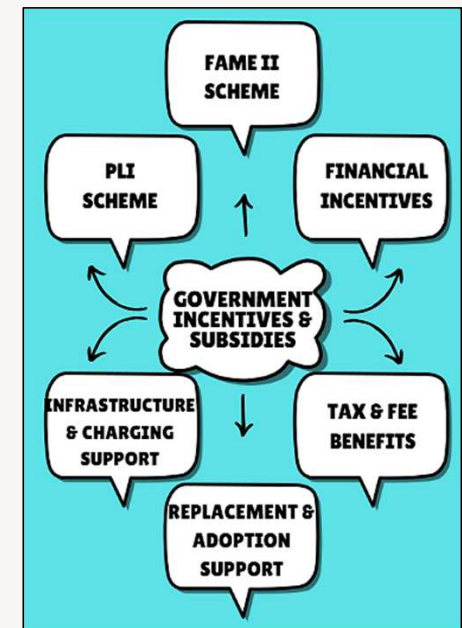
2. How do government incentives and subsidies impact the adoption rates of 2-wheelers and 4-wheelers?
Which states in India provided most subsidies?

Impact on 2-Wheeler EVs

- ✓ FAME II subsidies significantly reduced the upfront cost of 2W EVs, improving affordability
- ✓ State incentives such as tax benefits and financial support made EVs more accessible
- ✓ Overall demand was highly subsidy-driven, with incentives playing a key role in growth
- ✓ 2W EVs became price-competitive with petrol vehicles, driving higher adoption among urban users

Impact on 4-Wheeler EVs

- ✓ FAME II and state policies helped reduce upfront cost, but the impact was limited due to higher vehicle prices
- ✓ Subsidies lowered entry barriers but did not significantly affect overall affordability
- ✓ Adoption was driven more by total cost of ownership, fuel savings, and better product availability
- ✓ Overall, subsidies played a supporting role rather than directly driving demand



Secondary Research Questions

2. How do government incentives and subsidies impact the adoption rates of 2-wheelers and 4-wheelers? Which states in India provided most subsidies?

The table presents the leading 8 states offering the highest EV subsidies



Leaders in EV Subsidy Provision:

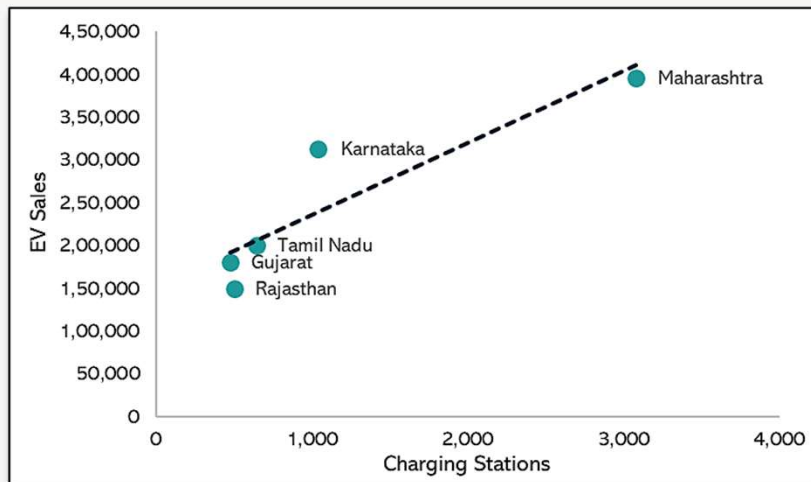
- ❖ **Gujarat, Maharashtra, and Delhi** emerged as key leaders by offering high subsidies, tax exemptions, and strong policy support, significantly boosting EV adoption.
- ❖ **Assam, Bihar, and West Bengal** also provided attractive financial incentives, including full road tax and registration fee exemptions, improving affordability
- ❖ **Karnataka and Tamil Nadu** focused more on infrastructure development and manufacturing incentives, supporting long-term growth of the EV ecosystem.

STATE WISE SUBSIDY FOR ELECTRIC VEHICLE				
State/Region	Subsidy per kWh	Maximum Subsidy 4-Wheeler	Maximum Subsidy 2-Wheeler	Road Tax Discount
Maharashtra	₹5,000	₹2,50,000	₹25,000	100%
Gujarat	₹10,000	₹1,50,000	₹20,000	50%
Delhi	₹5,000	-	₹30,000 (battery > 5 kWh)	-
Meghalaya	₹10,000	₹60,000	₹20,000	100%
Assam	₹10,000	₹1,50,000	₹20,000	100%
Bihar	₹10,000	₹1,50,000	₹20,000	100%
West Bengal	₹10,000	₹1,50,000	₹20,000	100%
Odisha	₹5,000	₹1,00,000	₹5,000	100%

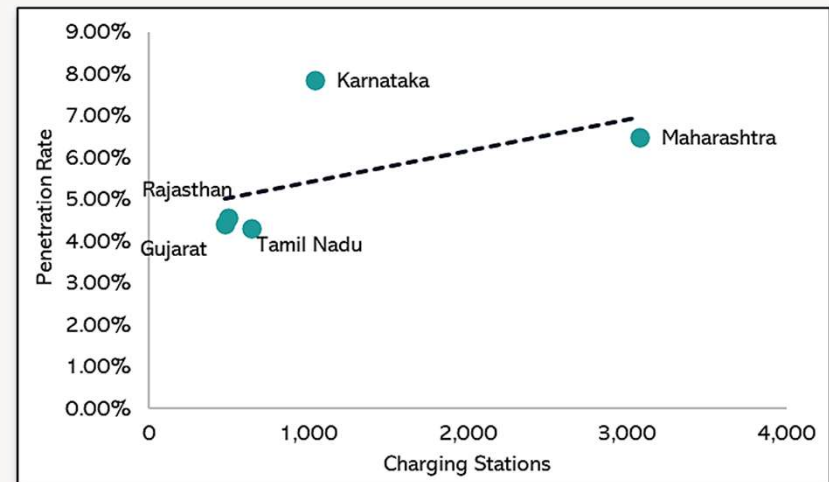
Source: e-vehicleinfo.com, ackodrive.com, ev2030.org, carbike360.com

Secondary Research Questions

3. How does the availability of charging stations infrastructure correlate with the EV sales and penetration rates in the top 5 states?



Correlation, $r = 0.905$



Correlation, $r = 0.519$

- The correlation between charging stations and EV sales ($r = 0.905$) shows a strong positive relationship, meaning states with more charging stations tend to have higher EV sales. This shows that charging infrastructure plays an important role in increasing EV sales.
- On the other hand, the correlation between charging stations and penetration rate ($r = 0.519$) is moderate, which means infrastructure helps, but other factors like government policies, pricing, and consumer preferences also affect EV adoption.

Secondary Research Questions

4. Who should be the brand ambassador if AtliQ Motors launches their EV/Hybrid vehicles in India and why?



Image Source: baltana.com

I believe that **MS Dhoni** would be an ideal brand ambassador for AtliQ Motors. He is one of the most respected and widely recognized personalities in India, with a strong fan base across all age groups. Known as 'Captain Cool,' he represents trust, reliability, and calm decision-making—qualities that align well with an automobile brand. His passion for bikes and automobiles further makes him a natural and authentic fit, helping the brand build strong connection and credibility.

Secondary Research Questions

5. Which state of India is ideal to start the manufacturing unit? (Based on subsidies provided, ease of doing business, stability in governance etc.)

In my opinion, **Gujarat** is the most suitable state to set up an EV manufacturing unit due to the following reasons:

1. Strong Subsidy Support 💰

Attractive incentives reduce overall manufacturing cost

2. Ease of Doing Business 📊

Fast approvals and investor-friendly policies

3. Lower Operational Costs ⚖️

More affordable land and labor compared to Maharashtra and Tamil Nadu

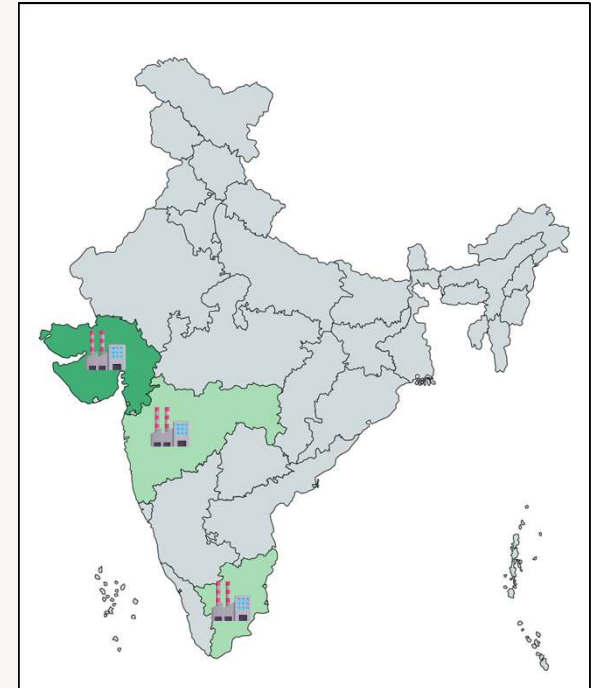
4. Growing Industrial Ecosystem 🏭

Expanding auto & EV manufacturing base

5. Strategic Location 📍

Strong port connectivity and logistics network

Alternatives : **Maharashtra** and **Tamil Nadu** follow as strong alternatives, offering superior infrastructure and manufacturing capabilities, though with relatively higher costs or slightly lower incentives.



Note: The data and insights presented in this chart are based on information from Wikipedia's Ease of Doing Business rankings, CarBike360's state-wise electric vehicle subsidy list, and the CareEdge 2023 State Ranking report and the Map image edited using Mapchart.net

Secondary Research Questions

6. Your top 3 recommendations for AtliQ Motors.



Market Entry

1. Focus on **2W EV segment** (higher demand & penetration compared to 4W)
2. Focus on **budget-friendly EVs** for middle-class consumers to drive mass adoption
3. Once strong brand trust is built, the company can strategically enter the 4W EV segment.



Manufacturing

Set up in Gujarat (High subsidies + Ease of doing business + Lower operational costs)



Expansion

Phase 1:
Start with Maharashtra, Karnataka & Gujarat
Phase 2: Expand to Tamil Nadu
Phase 3: After achieving good demand and brand presence, enter emerging markets like Chhattisgarh and Goa



Timing & Ecosystem

- Partner with **charging networks**
- Launch in **Q1 & Q4** (festive seasons + policy incentives)



Thank you

| Saranya Koduru 